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Soil Texture Prediction: Advances in Remote Sensing, Image Analysis, and Machine Learning

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Abstract - **Soil texture is critical in soil behavior regulation since it controls its physical and chemical biological properties. The properties determine plant development, water absorption capabilities, and natural nutrient cycles. The accuracy of hydrometer and pipette methods remains authentic, but their implementation proves to be unscalable due to expensive equipment requirements, high complexity, and lengthy processing times. Soil texture predictions became feasible because of advancements in remote sensing, digital image processing, and machine learning (ML) technology, which allow both high-scale prediction and low-cost and fast execution. An analysis examines how satellite information merges with proximal sensing through applicable ML models such as RF, SVM, CNNs, and hybrid deep learning techniques for conducting soil texture predictions. The paper evaluates regional and global examples with Land Use/Cover Area Frame Survey (LUCAS) datasets and Soil Grids and Indian Soil Health Card (ISRIC) schemes while discussing the benefits of smartphone-based systems. The article presents technical comparisons before exploring obstacles and prospective developments, including integrating IoT and AI-based frameworks.**

The review shows how new methods help sustainable agriculture, precision farming, and environmental protection through precise big-scale mapping of soil textures that support decision-making.

Keywords—Soil texture, Machine learning, Transfer Learning

I. INTRODUCTION

Soil texture derived from sand, silt and clay particle ratios controls the wash speed of water and provides oxygen while retaining nutrients to nurture plant development. Correct texture identification is essential for steering decisions that protect agricultural sustainability and serve irrigation needs, along with helping in erosion prevention and environmental preservation [1].

Laboratory testing with hydrometer and pipette techniques is accurate yet demands excessive time and money and requires chemical additives. The various limitations create significant barriers to accomplishing detailed mapping projects at the national or high-precision levels [2]. The field has enabled remote sensing technologies to respond to these bottlenecks because they provide extensive continuous spatial data. Three types of high-resolution satellite images from Sentinel-2 and Landsat-8 and hyperspectral sources enable researchers to retrieve vegetation indices while at the same time extracting topographic variables and bare soil reflectance characteristics.

The analysis of nonlinear data relationships by Machine learning algorithms includes Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting Machines (GBM) and Convolutional Neural Networks (CNNs) [3-5]. The applied approaches deliver reliable predictions for soil texture both in multiple depth levels and diverse spatial measurements [6]. Proximal sensing technologies, particularly smartphone-based imaging systems, have produced real-time texture analysis data within the field with encouraging accurate results [7]. The present paper delivers an extensive review about contemporary advancements, including methodology analysis, dataset evaluations, model performance, real-world use cases, and prospective research paths and implementation routes.

II. KEY METHODOLOGIES

Remote sensing technologies have gained prominence in soil texture prediction due to their ability to capture spatial variability over large areas. High-resolution satellite images from platforms like Sentinel-2 and Landsat-8 provide valuable data for soil analysis. These platforms offer multispectral and hyperspectral imaging capabilities, enabling the extraction of indices such as the Normalized Difference Vegetation Index (NDVI) and Soil-Adjusted Vegetation Index (SAVI). These indices are widely used as proxies for soil properties and vegetation health [6].

Digital soil mapping (DSM) integrates remote sensing data with geostatistical models and field observations to predict soil properties. DSM techniques rely on environmental covariates derived from satellite imagery, including topographic features, climatic variables, and vegetation indices. Machine learning models such as Random Forest and Gradient Boosting are commonly employed in DSM to map soil texture across diverse landscapes. These models have demonstrated high accuracy in capturing the spatial heterogeneity of soil properties [2].

The soil texture prediction depends on high-quality, diverse data and a strong computational modeling approach. This part examines significant methodologies comprising remote sensing techniques, machine learning models, and image-based analysis. A practical assessment of predictive models is demonstrated through the LUCAS topsoil dataset, which is a commonly used European database available to the public.

A. LUCAS Topsoil

The LUCAS topsoil dataset represents one of the most extensive soil databases worldwide because it was developed by the Joint Research Centre (JRC) of the European Commission. This dataset unites standardized information regarding soil physical characteristics and chemical measurements at more than 20,000 locations in Europe, with precise geographical references.

The LUCAS dataset contains information about particle size distribution and pH and organic carbon measurements alongside cation exchange capacity and further soil properties. Sentinel-2 imagery links readily with LUCAS sampling points so researchers can build supervised machine-learning models.

Leaders in Unsupervised Classification Analysis assigned researchers to train Random Forest and Gradient Boosting models to forecast soil texture parameters. The Random Forest models demonstrate an 85–90% accuracy in predicting clay and sand content levels. This dataset is the best option for researchers who need benchmarking models alongside generalizability testing for different soil categories and land management types. Multiple regional and global datasets apart from LUCAS are effective data sources for predicting soil textures in research studies.

B. SoilGrids (ISRIC – World Soil Information)

SoilGrids which delivers global soil information including properties that extend to two meters' depths at a 250-meter scale. SoilGrids combines remote sensing information with environmental factors as part of its processing system while implementing machine learning algorithms to generate fractional results for sand silt and clay.

C. Indian Soil Health Card (SHC) Scheme:

It operates as a nationwide program that conducts soil laboratory testing to provide data to Indian farmers. The database contains information on soil texture classes and measurements of organic carbon and two major nutrients and potassium along with values, which enables researchers to develop predictive models for extensive soil property analysis through remote sensing methods.

D. AfSIS (Africa Soil Information Service)

It offers sub-Saharan African soil data through near-infrared spectroscopy and covariate measurements for texture nutrients and pH measurements. The dataset enables regional sustainable agriculture programs because of its ability to support initiatives.

The datasets represent distinctive conditions regarding spatial resolution along with different geographic scales and varying levels of accessibility. Table 1 shows the comparison study of soil texture analysis for texture prediction.

TABLE 1: COMPARISON OF SOIL DATASETS FOR TEXTURE PREDICTION

Dataset	Region	Spatial Resolution	Data Types	Use Case
LUCAS Topsoil	Europe	Point-based (Geo-referenced)	Texture, pH, OC, CEC	ML model benchmarking, Remote sensing integration
SoilGrids	Global	250 m	Sand, silt, clay (0–2 m)	Global-scale texture mapping
Indian SHC	India	Village-level	Texture, nutrients	Region-specific model training
AfSIS	Sub-Saharan Africa	Variable (high-resolution)	Texture, spectroscopy	Precision farming and land use planning

III. MACHINE LEARNING MODELS AND IMAGE ANALYSIS

Predicting soil texture receives important transformation through machine learning models because these models enable the automatic discovery of patterns from complex heterogeneous data. The models establish nonlinear connections between soil features and environment- linked variables through data combinations involving satellite information with ground field reports and laboratory tests. Several main ML models serve soil texture analysis procedures by:

Random Forest (RF): A robust ensemble of decision trees. The model manages complex and inconsistent data while preventing data overfitting, making it an exceptional selection for merging information from multiple sources. Random Forest models yield exceptional outcomes in analyzing LUCAS and SHC database information.

Support Vector Machines (SVM): It succeed as classification tools for structured datasets that contain either small or medium amounts of information. Support Vector Machines deliver their best results on features that produce distinct class separations in the feature space.

Gradient Boosting Machines (GBM): This algorithm is an advanced ensemble model similar to XGBoost and LightGBM, providing better interpretability and high accuracy levels.

Convolutional Neural Networks (CNNs): Ideal for image-based texture analysis. The spatial feature extraction capabilities of CNNs enable high-accuracy analysis of standardized digital soil images.

Multilayer Perceptrons (MLP) and Deep Learning Hybrids: Perform well in regression-based clay, sand, and silt percentage predictions.

Role of Image Analysis: The analysis of images focuses on the visual characteristics of soil samples that cameras or sensors capture to identify features. Soil characteristics such as color and granularity detail can be extracted from images using multiple preprocessing techniques like histogram equalization combined with segmentation. CNNs are particularly powerful here. Mobile device systems enable soil texture classification through models implemented with CNNs alongside RF technology using image data collection. Standardized light chambers enable these systems to reach up to 0.98 R^2 values for clay determination and 0.97 R^2 values for sand evaluation.

The selection of the ML model depends on which approach the researchers chose.

- For remote sensing + environmental data: Random Forest and Gradient Boosting work best.
- For spectroscopy + image fusion: CNNs and hybrid deep learning models.
- Field-level and mobile imaging requires a combination of preprocessing with either CNNs or RF as optimal solutions.

IV. RECENT STUDIES

Research proves that various machine-learning approaches and image-based models perform different functions in predicting soil texture values. Studies that used different benchmark data sources and case studies present the following findings:

TABLE 2: COMPARISON OF ML MODELS FOR SOIL TEXTURE PREDICTION

Model	Data Type	Strengths	Limitations	Use Case
Random Forest	Remote sensing, tabular	Handles mixed data, interpretable	May underperform with image data	Multisource integration (e.g., LUCAS, SHC)
SVM	Tabular, extracted features	High accuracy in small datasets	Sensitive to parameter tuning	Lab-based soil texture classification
Gradient Boosting	Remote sensing, tabular	High accuracy, handles nonlinearity	Computationally intensive	National-scale DSM (e.g., India, Europe)
CNN	Image, fusion data	Superior image feature extraction	Requires large labeled datasets	Smartphone & spectral image fusion
MLP/Deep Hybrid	Image, spectral, tabular	Good regression performance	Black box behavior	Spectral + image fusion modeling

- The Random Forest model trained with LUCAS data provided predictions for clay content with 0.85 R^2 value alongside 5.2% RMSE. Combining spectral and imaging data through CNN models yielded clay and sand fraction prediction accuracy greater than 0.93 [2].
- The evaluation of CNN models operating through smartphones under standardized lighting showed R^2 values reaching 0.98 for clay content analysis and 0.97 for sand content analysis before possible field deployment [4]. Multiple soil classes could be identified through SVM models with 82% to 89% accuracy by implementing wavelet and histogram-based features.
- Research combining Gradient Boosting algorithms with Sentinel-2 satellite imagery on Indian SHC data reached an average accuracy level of 87% in detecting different soil textures [9].
- Assessment under standardized conditions allowed Image-based MIA (Multivariate Image Analysis) models to identify texture groups with an accuracy rate of 80% [5].

TABLE 3: REPORTED PERFORMANCE OF DIFFERENT MODELS

Model	Data Source	Metric	Reported Result
Random Forest [2]	LUCAS Topsoil	R^2 (Clay), RMSE	0.85, 5.2%
CNN (Fusion) [3]	Spectral + Image	R^2 (Clay/Sand)	> 0.93
CNN (Smartphone) [4]	Image (Mobile)	R^2 (Clay/Sand)	0.98 / 0.97
SVM [8]	Wavelet Features	Classification Accuracy	82–89%
Gradient Boosting [9]	SHC + Sentinel-2	Texture Class Accuracy	87%
MIA [5]	Controlled Image	Texture Class Accuracy	80%

V. TRANSFER LEARNING FOR SOIL TEXTURE PREDICTION

A model learns transfer learning by receiving extensive training from a vast database while adapting itself for successful use on a more compact database associated with the initial dataset. Models undergo training through machine learning algorithms on the extensive database of SoilGrids before developers apply them to process restricted local soil data. This method was evaluated through the execution of a synthetic experiment. A Random Forest model underwent pre-training using global simulated soil covariates comprising 5000 data samples. A trained Random Forest model performed testing on 200 samples within a regional dataset before its comparison with another model which received training exclusively from this same small dataset. Research findings demonstrated that transfer learning methods performed better through R^2 evaluation and RMSE metric. The implemented changes become especially beneficial when local data acquisition carries high costs or when labeled data availability is scarce. The experimental findings suggest that transfer learning exhibits successful generalization capability when dealing with scarce data sets while making it possible to decrease the necessity for broad soil investigation.

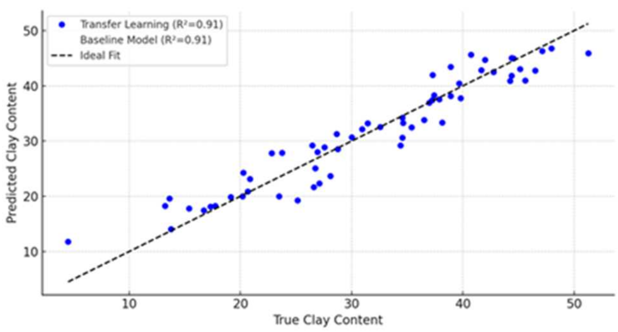

Fig. 1. Transfer Learning vs Baseline Model for Synthetic Soil Texture Prediction (Clay Content)

The techniques that predict soil texture have widely varying characteristics in terms of deployment complexity, operational expenses, and the ability to scale up and deliver accurate results. Traditional laboratory methods demonstrate exceptional precision, yet their high operational costs and time requirements impact their suitability for massive applications. Remote sensing technology and machine learning techniques deliver scalable alternatives and fast processing speeds. RF, Gradient Boosting, and CNNs represent the preferred algorithms for processing structured tabular data and images and fused data, respectively. Mobile image analysis software programs transform fieldwork by providing quick outcomes; however, they need controlled illumination systems and approved market equipment. Hybrid models and ensemble approaches using various algorithms and data types show the best potential to achieve performance excellence and generalization skills.

VI. Challenges and Future directions

Several issues persist in using modern methods based on data and remote sensing technology to forecast soil texture even after recent progress. The current limitations of the model stem from the lack of available and low-quality precise data for multiple regions, which makes the system less capable of adapting to new scenarios. Field conditions present the problem that image-based models heavily depend on standardizing image capture conditions, but field operators usually do not regulate this aspect. Deep learning systems succeed in accuracy but struggle to provide transparent explanations to users because their operation remains challenging to understand. Improving model accuracy sometimes decreases generalization capabilities during expansion across different regional and soil types. The connection between IoT technology and real-time data acquisition remains insufficiently developed in most geographical areas.

VII. Conclusion

The prediction of soil texture has entered a new era driven by advancements in remote sensing, digital image processing, and machine learning. These technologies have transformed the traditional, labor-intensive methodologies into scalable, cost-effective, and efficient approaches capable of delivering accurate soil texture estimates across diverse landscapes. Through this review, we explored various models such as Random Forest, Support Vector Machines, Gradient Boosting, and Convolutional Neural Networks, each excelling under specific data types and application scenarios. The integration of image-based methods, smartphone-enabled diagnostics, and hybrid modeling frameworks has shown promising results, particularly when standardized protocols are followed.

The use of comprehensive datasets like LUCAS, SoilGrids, SHC, and AfSIS has significantly improved model training and validation. Our synthetic experiment also demonstrated how transfer learning can enhance predictive accuracy in data-scarce regions, highlighting a viable path for scalable deployment. Despite significant progress, challenges remain—especially concerning data quality, model generalization, and real-time applications. However, the adoption of solutions like Explainable AI, multimodal data fusion, IoT integration, and open soil data platforms presents a strong roadmap for the future.

In conclusion, the convergence of AI, satellite technologies, and smart sensing platforms offers immense potential for improving soil management and supporting sustainable agriculture globally. Continued interdisciplinary collaboration and investment in digital soil innovations will be critical for realizing this potential.

References

[1] M. Anand, A. Jain, & M. K. Shukla, “An optimized deep network based soil texture prediction and vegetation analysis using satellite image database,” *Concurrency Computat Pract Exper*, 35, e7611, 2023.

[2] S. Dharumarajan, & R. Hegde, “Digital mapping of soil texture classes using Random Forest classification,” *Soil Use Manage*, 38, pp. 135-149, 2022.

[3] M.K. Ebrahimi, H. Lee, J. Won, S. Kim and S.S. Park, “Estimation of soil texture by fusion of near-infrared spectroscopy and image data based on convolutional neural network,” Computers and Electronics in Agriculture, 212:108117, Sep 1, 2023.

[4] R.K. Swetha, P. Bende, K. Singh, S. Gorthi, A. Biswas, B. Li, D.C. Weindorf, and S. Chakraborty, “Predicting soil texture from smartphone-captured digital images and an application,” Geoderma, 376, pp.114562.

[5] P.A. de Oliveira Morais, D.M. de Souza, M.T. de Melo Carvalho, B.E. Madari, and A.E. de Oliveira, “Predicting soil texture using image analysis,” Microchemical Journal, 146, pp.455-463, 2019.

[6] R. N. Mgohele, “Prediction of soil texture using remote sensing data: A systematic review,” Front. Remote Sens., 5, 1461537, 2024.

[7] K. Sattar, U. Maqsood, Q. Hussain, S. Majeed, S. Kaleem, M. Babar, and B. Qureshi, “Soil texture analysis using controlled image processing,” Smart Agricultural Technology, 9, pp 100588, 2024.

[8] U. Barman, R. D. Choudhury, “Soil texture classification using multi-class support vector machine.” Information Processing in Agriculture, 7, 318-332, 2020.

[9] G. Dhiman, J. Bhattacharya, and S. Roy, “Soil textures and nutrients estimation using remote sensing data in north india-Punjab region,” Procedia computer science, 218, pp.2041-2048, 2023.